

## AN2DL - Second Challenge Report

### TeamSevilla

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## 1 Introduction

This project focuses on image classification using **convolutional neural networks** (CNNs). The goal is to classify images of diseased tissue samples into the correct diagnosis.

## 2 Problem Analysis

We first examined the dataset by analyzing the class distribution. The triple negative class is significantly underrepresented, with 349 samples compared to roughly 1,000 samples for each of the other classes. This class imbalance is expected to pose challenges during training, potentially biasing the model toward the majority classes.

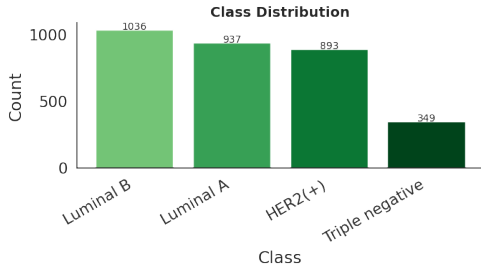


Figure 1: Class distribution in the training dataset

We then performed a qualitative inspection of individual data samples. This analysis revealed

the presence of noticeable artifacts in the training dataset. In particular, we identified two distinct types of recurring artifacts: images containing Shrek-like patterns and images characterized by green slime-like textures.



Figure 2: Examples of artifact type 1: "Shrek"



Figure 3: Examples of artifact type 1: "Slime"

These artifacts may introduce spurious correlations and negatively impact the model's ability to learn clinically relevant features, and therefore must be carefully considered in subsequent preprocessing and modeling steps.

## 3 Method

To perform classification, we initially trained a convolutional neural network (CNN) using the RGB

image as input, augmented with the corresponding grayscale mask as a fourth channel. This configuration served as our baseline model (see Table 1). The inclusion of the mask channel was motivated by the need to explicitly guide the network toward the region of interest.

Then, we made modifications by experimenting with different techniques to improve the model, as explained in the following section.

For most of our models we used a stratified 80% test / 20% validation data split.

## 4 Experiments

**Regularization and loss reweighting.** Early experiments revealed clear signs of overfitting in the baseline model. To mitigate this issue, we systematically tuned the dropout rate and concluded that the best result was obtained with dropout rate 0.3, being an improvement over the baseline, as seen on Table 1. In a subsequent set of experiments, we replaced the standard cross-entropy loss in Equation 1 with a weighted cross-entropy loss to account for class imbalance, particularly the under-representation of the triple negative class.

- **Cross-Entropy loss function**

$$-\sum_{n=1}^N \mathbf{t}_n^T \log g(x_n | w) \quad (1)$$

- **Weighted Cross-Entropy loss function**

$$-\sum_{n=1}^N \sum_{c=1}^K \alpha_c t_{n,c} \log g_c(x_n | w) \quad (2)$$

$$\alpha_c = \frac{N}{K \cdot N_c} \quad (3)$$

**Pretrained architectures.** We then explored transfer learning by fine-tuning pretrained architectures, specifically ResNet-18, DenseNet121, MobileNetV3 and Virchow2. These models achieved improved generalization performance compared to the vanilla CNN and are reported in Table 1.

**Artifact-aware modeling.** Finally, we addressed the challenge posed by dataset artifacts by explicitly modeling their presence. We implemented two

dedicated CNNs, one for each artifact type (Shrek-like patterns and green slime). These networks were trained on a reduced subset of affected samples, augmented with images of the artifacts themselves to create a synthetic training set (available here).

The trained artifact detectors were then used to enrich the original dataset by adding two binary flags—one per artifact—to the labels dataframe. We subsequently explored several approaches:

- **Six-channel input:** Train the model on a six-channel input composed of the RGB image, the grayscale mask, and the two artifact flags.
- **Late fusion:** Modify the architecture to use a four-channel input (RGB + mask) and incorporate the artifact flags into a final MLP for classification.
- **Augmentations:** Apply both Virchow2 model-specific transformations and additional random augmentations. Also other techniques like CutMix and MixUp were tried.

We evaluated these approaches using both a vanilla CNN and a pretrained backbone; the corresponding results are reported in Table 1.

## 5 Results

Our best results were taken on the models:

- **Fine-tuned late-fusion transfer learning model based on Virchow2 [1].** Since Virchow2 was pretrained on whole-slide images (WSIs), input images were resized to 224×224 pixels and processed using the default Virchow2 normalization. To improve robustness, additional data augmentations—including random rotations and subtle jittering—were applied during training. The architecture is illustrated in Figure 4.
- **Fine-tuned late-fusion transfer learning model based on Virchow2 with an augmented training set (best-performing).** This model follows the same architecture as the previous one, but additionally leverages the previously mislabelled dataset from Challenge 2 by inferring pseudo-labels. Specifically, the model was used to predict class probabilities for these samples, and only those as-

Table 1: Our models. Best results are highlighted in **bold**.

| Model                                                                 | Validation F1 | F1 on Test    |
|-----------------------------------------------------------------------|---------------|---------------|
| Baseline (Vanilla CNN, 4ch: RGB + Mask)                               | 0.4148        | 0.2846        |
| Dropout_tuned (dropout_rate=0.3)                                      | 0.3742        | <b>0.3441</b> |
| ResNet-18                                                             | 0.4637        | 0.2605        |
| Baseline augmented (hflips, ColorJitter, Rotations, etc.)             | 0.368         | 0.2267        |
| Baseline with 3 conv blocks                                           | 0.391         | 0.3223        |
| Focus filtered image + mask                                           | 0.3602        | 0.284         |
| Deeper CNN                                                            | 0.3353        | 0.1944        |
| Baseline with dropout rate 0.3 and $64 \times 64$ resize              | 0.3293        | 0.2755        |
| Transfer Learning with MobileNetv3                                    | 0.4881        | 0.2874        |
| Transfer Learning with DenseNet121                                    | 0.4294        | 0.2789        |
| artifacts_DenseNet121_6ch (RGB + mask + Artifacts flags as channels)  | 0.4544        | 0.2687        |
| artifacts_DenseNet121_latefusion (4ch: RGB, mask + 2 artifacts flags) | 0.5164        | 0.3005        |
| artifacts_latefusion                                                  | 0.318         | 0.2478        |
| artifacts_Virchow2_latefusion                                         | 0.5529        | <b>0.3441</b> |
| artifacts_Virchow2_latefusion+mask                                    | 0.5182        | 0.3375        |
| artifacts_Virchow2_latefusion_GAN_augmented                           | 0.5042        | 0.3389        |
| artifacts_Virchow2_latefusion+CutMix                                  | 0.5352        | 0.2996        |
| artifacts_Virchow2_latefusion-OLD_DATASET_AUGMENTED                   | 0.7815        | <b>0.3599</b> |

signed with the highest confidence were selected. These high-confidence samples were then used to augment the Challenge 2 v2 training set, and the model was retrained under the assumption that the inferred labels were correct.

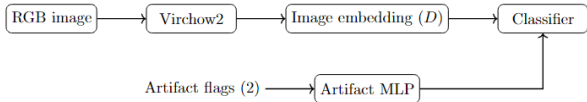


Figure 4: Architecture with late fusion of artifact metadata.

## 6 Discussion

The initial modeling efforts using Vanilla CNN confirmed clear signs of overfitting, marked by a large gap between Validation F1 and Test F1. The best model achieved the highest performance benchmark as seen on Table 1. This validates that explicitly integrating artifact metadata with high-level visual features (Virchow2) is the most effective architectural approach. Despite this success and the use of Weighted Loss, the overall low Test F1 suggests that the class imbalance, particularly the underrepresentation of the Triple Negative class, remains the

dominant limitation. Furthermore, we have encountered a limited impact of data augmentation, possibly because these methods created overly modified images that the specialized Virchow2 model could not correctly interpret.

## 7 Conclusions

We have shown that a specialized, metadata-informed architecture is required to effectively address data artifacts and class imbalance. The Artifacts-Virchow2 late-fusion model is confirmed as the most stable and highest-performing configuration, combining transfer learning with explicit metadata integration.

Future work should also explore alternative pre-trained backbones specifically trained on whole-slide images, such as MahmoodLab/UNI, and further target the minority-class performance bottleneck through strategies like Hard Sample Mining that prioritize high-loss triple-negative samples.

## References

- [1] Paige AI. Virchow2: Paige AI Histopathology Model. <https://huggingface.co/paige-ai/Virchow2>, 2023. Accessed: 2025-12-16.